Swati Aggrawal

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PROFESSIONAL SUMMARY

MSc Computer Science graduate from Sapienza University of Rome specializing in Artificial Intelligence and Machine Learning. Experienced in developing AI-enabled applications, machine learning pipelines, and backend services using Python, FastAPI, PyTorch, Docker, and cloud technologies. Hands-on experience with Generative AI, NLP, Computer Vision, REST APIs, and deploying scalable AI solutions. Familiar with Java, frontend development, databases, and software engineering practices. Interested in building production-ready AI applications.

SKILLS

- **Programming :** Python, Java, SQL, C++, Bash
- **AI/ML :** Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, Generative AI, Large Language Models (LLMs), Transformer Models, Model Evaluation & Optimization, OpenCV
- **Backend & APIs:** FastAPI, RESTful APIs, Backend Development, API Integration, Microservices Fundamentals
- **Frontend :** React.js (Basic), HTML, CSS, JavaScript
- **Databases :** Relational Databases, PostgreSQL, MySQL
- **Cloud & DevOps :** Docker, AWS, Cloud Computing Fundamentals, Git, GitHub
- **Software Engineering :** Object-Oriented Programming (OOP), Data Structures & Algorithms, Unit Testing, Agile/Scrum, Software Development Lifecycle (SDLC)

EDUCATION

Sapienza Università di Roma **May 2026**

Master of Science in Computer Science

- **Relevant Coursework:** Computer Vision, Deep Learning and Applied AI, Natural Language Processing, Machine Learning, Distributed Systems
- **Thesis:** Evaluating Pretrained Vision Transformers Under Continual Learning.
- Selected for a competitive international master's program.

Sharda University **July 2023**

Bachelor of Technology - CSE with AI and ML

- **Relevant Coursework:** Operating Systems, Data Structures, Machine Learning, Databases, Cloud Computing
- **Thesis:** Virtual Generative Model Inducing Human Identity to Computers in Metaverse
- Awarded **merit-based scholarship** (Top 5–10% of cohort).

PROJECTS

DocDelta AI-Powered Document Intelligence & Comparison Platform

- Developed an AI-powered document intelligence application with backend APIs for automated document analysis and semantic comparison.
- Designed RESTful endpoints using FastAPI for automated document analysis and change tracking workflows.
- Developed backend workflows for document processing and semantic comparison.
- Integrated transformer-based ML models into application pipelines for automated inference.
- Implemented semantic matching to identify added, removed, and modified content across document versions.
- Containerized the application using Docker and deployed it on AWS ECS Fargate.
- Configured application monitoring and logging using AWS CloudWatch.
- **Technologies:** Python, FastAPI, Sentence Transformers, Docker, AWS ECS Fargate, CloudWatch, REST APIs

PosePal Real-Time Yoga Pose Classification System

- Developed a real-time computer vision application that classifies yoga poses from live webcam input.
- Built and integrated a CNN-based image classification model for multi-class pose recognition.
- Implemented image preprocessing, normalization, and data augmentation techniques to improve model performance.
- Integrated model inference into a lightweight web application for real-time predictions.
- Applied frame-level prediction smoothing to reduce prediction fluctuations during live inference.
- Developed a lightweight web interface for real-time webcam-based pose classification.
- **Technologies:** Python, TensorFlow, Keras, OpenCV, Scikit-Learn

Metal Defect Detection using Computer Vision

- Developed computer vision models for automated defect detection using CNN, SVM, and Random Forest.
- Built preprocessing, training, and evaluation pipelines.
- Compared deep learning and traditional ML approaches.
- Performed model evaluation using accuracy, confusion matrices, and performance analysis.
- **Technologies:** Python, TensorFlow, Keras, OpenCV, Scikit-Learn

RESEARCH EXPERIENCE

Continual Learning for Computer Vision (Master's Thesis) — Nov 2025

- Evaluated pretrained Vision Transformer architectures under Task-Incremental learning settings.
- Designed experimental pipelines for training, evaluation, and comparison of computer vision models across sequential tasks.
- Evaluated model performance under changing data distributions and analyzed model reliability.
- Investigated catastrophic forgetting and knowledge retention in transformer-based visual learning systems.
- Developed reproducible training and evaluation pipelines using Python and PyTorch.
- Automated experiment tracking and performance benchmarking across multiple continual learning scenarios.
- Documented experimental findings and identified research opportunities for future development of continual learning systems.
- **Technologies:** Python, PyTorch, Vision Transformers, Computer Vision

Machine Learning for Air Pollution [IEEE Publication] — May 2023

- Published IEEE conference paper evaluating machine learning approaches for environmental prediction and monitoring.
- Evaluated supervised ML algorithms including Random Forest, SVM, Decision Trees, and Neural Networks across environmental datasets.
- Performed comparative analysis using model performance metrics to assess accuracy, robustness, and generalization.
- Presented scientific findings through structured research documentation and publication.